

Fused Photo: Augmented Reality Staging for Photorealistic Visualization in Online Home Retail

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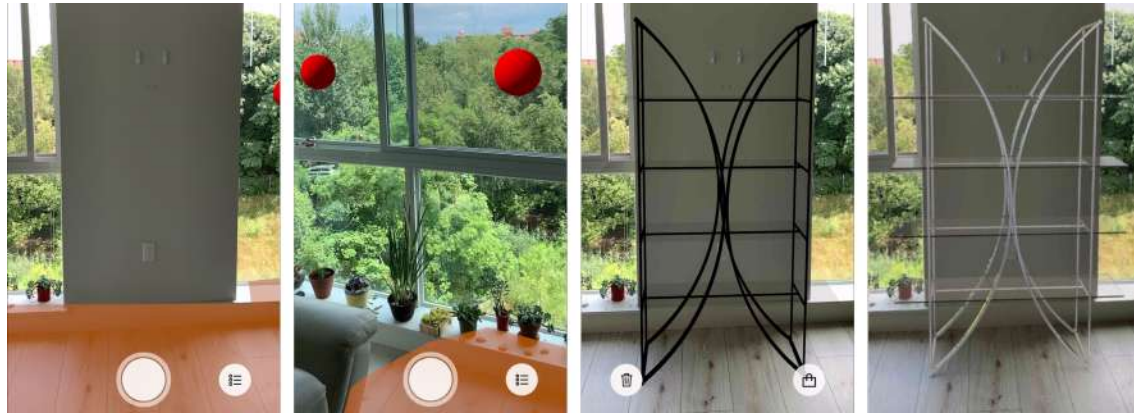


Figure 1: With Fused Photo, users can capture a picture; detect planes and light sources; place a low-res real-time model in the scene; and render a photorealistic version that incorporates the scene lighting.

ABSTRACT

Retail experiences have started adopting Augmented Reality (AR) as a tool to let customers see products in their space before committing to a purchase. Although online retailers use high resolution 3D models to create onsite imagery, limitations in the rendering capabilities of phones require them to optimize models for real-time use, diminishing the visual quality of the experience and widening the gap between what the customer thinks they are buying and what they get. We present Fused Photo, a system that leverages AR to capture the customer's spatial context and recreate it in a remote rendering environment that then composites photorealistic images of products onto their space. Results show that Fused Photos outperform other AR visualizations in the market both in realism and ability to represent a product's appearance, while being comparable in quality to photorealistic renders staged manually by artists.

CCS CONCEPTS

• **Applied computing** → **Online shopping**; • **Computing methodologies** → **Rendering**; **Mixed / augmented reality**.

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KEYWORDS

Augmented Reality; Online Retail; Rendering

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1 INTRODUCTION

Research has shown that the use of Augmented Reality (AR) in retail enhances the user's experience by providing accurate product information and meaningful context, leading to reduced user anxiety and facilitated decision making [36]. The online retail industry has been an early adopter of AR both as a means to show customers how products will look in context [6] and as a marketing medium to increase brand awareness [5, 38]. AR experiences in retail are prominent in segments where the spatial dimensions and the visual appeal of a product are important to the customer such as: wearable products (e.g. watches [43], footwear [47], and glasses [48]), cosmetics [33], and home goods [7, 31, 35].

In the online home retail space, AR View-in-Room features improve the user experience by letting customers know if a product will fit within the context of furniture they already own as well as in their space, at real-world scale [3, 44, 46]. These features are predominately available to consumers on their mobile phones [18]. However, mobile phones are resource limited devices that require lower level of detail (LOD) 3D content in order to sustain real-time

experiences [8, 37]. These low LOD models can result in visualizations that may not accurately portray how products will look to the customer when they ship to their homes. We believe high quality, photorealistic visualizations in AR can improve the user experience by reducing the gap between the virtual representation and the physical product, instilling more confidence in the customer's purchasing decision.

Online retailers have used high LOD 3D models of their products to produce photorealistic imagery, removing the costs of moving merchandise to centralized photo studios and hiring professional photographers to stage shots [11, 12]. However, in order to use high LOD models in real-time AR experiences, retailers must reduce their resolution to accommodate for low resource environments, such as a customer's phone. Designer services that render high LOD models in the user's space, such as Modsy [28], manually recreate an entire 3D scene to match the structural properties and lighting of the user's space. As a result, embedding of products in the customer's context at photorealistic quality can be expensive and time consuming.

AR platforms are increasingly capable of capturing context from the user's environment. Advances in computer vision and mobile device hardware allow these frameworks to capture lighting and recreate structural features of a space automatically [29]. In this paper, we present Fused Photo, a system that lets customers enhance the quality of their AR visualizations by using low resolution 3D models as proxies to recreate a higher LOD scene in an offline rendering workflow, producing photorealistic composites of synthetic products in the user's space.

2 RELATED WORKS

Compositing synthetic objects into real scenes is a longstanding goal of computer graphics [24]. A focus of this research consists of extracting information from the real image that can be applied to enhance the visual coherence between the synthetic object and the scene through differential rendering [10]. An example of this is estimating lighting from the image, like in DeepLight, [14, 15, 22, 40] and applying it to the synthetic objects in a technique known as image-based lighting [39].

Among the seminal works in photorealistic compositing is Fournier et al's system to insert imagery into video feeds [13]. This line of research has led to the creation of tools that facilitate compositing for 3D artists. For instance, Karsh et al extended Fournier's work by developing a drag-and-drop tool for object insertion into photos [23]. This system created a rough reconstruction of the scene from an LDR photograph whereupon the user could annotate light sources and add more refined scene geometry before inserting synthetic models into the scene to be rendered. Similarly, Adobe Dimension [1] streamlines the compositing process by using depth and light estimation to allow models to then be dragged and dropped on top of photographs. Our system extends the aforementioned body of work by leveraging AR as the tool that facilitates the extraction of context, which is oftentimes limited in a single image, while enabling in-situ rendering capabilities.

We frame the contribution of our work on two fronts. First, we posit AR as a tool that enables in-situ distributed rendering, acting as an intermediate tool to stage low-res models in the user's

context and recreate contextual information offline to produce photorealistic imagery. Second, we apply this technique in the domain of online home retail as an alternative to traditional visualization paradigms that trade-off quality for real-time operation, implementing a system that is of comparable quality to imagery offered in online staging design services.

2.1 Mixed Reality for In-Situ Rendering

Mixed reality has been employed as an intermediate staging tool in a number of domains because of its inherent provided contextualization. Within cinematic virtual production, it has been used to create in-situ lower-res visualizations of special effects and set extensions. Collaborative filmmaking tools [9, 17, 20, 25, 45] allow directors to understand the spatial layout of the scene with lightweight asset additions while on set, before VFX is composited in post-production. This is similar to the technique we propose, except it focuses only on the position of elements in the scene as opposed to incorporating context in the pursuit of photorealism.

Within the distributed rendering space for photorealistic mixed reality, Pomi et al's system focused on extending OpenRT for mixed reality composites by utilizing proxies as stand-ins for future high fidelity renders [34]. Though the usage of proxies and distributed rendering is similar to our system, our work differs by eliminating the need to infer the geometric composition of the scene and lighting by instead using AR's inherent capabilities to capture the needed topology. Additionally, we employ an in-situ staging approach as opposed to their offline methods.

Authoring tools focusing on photorealism in mixed reality include the Dreamhouse interior design system by Park et al [32]. Their tool requires a scan of the space via an RGB-D camera to retrieve scene topology as well as global illumination, which then acts as the foundation for accurately rendered and lit furniture, as seen through an HMD. This tool focused more on immersion and spatial interaction with objects; our system differs in providing no additional overhead for photorealism beyond the initial photo capture and minimal user annotation for lighting.

2.2 3D Visualizations in Online Home Retail

In the online home retail market, we consider a number of AR applications, including Wayfair View in Room [46], Ikea Place [21], Houzz View in Room 3D [19], and Amazon AR View [2]. These consumer-facing applications focus on in-context clarity of fit and placement of furniture [31], but provide non-photorealistic visual representations due to mobile hardware constraints.

Virtual design services, such as Modsy [28] create furnished 3D renderings from customer-provided photos. The output of services such as these can be combined with VR applications for proven increased presence and consumer lift, as seen in the study done by Brenner et al [4]. However, due to manual involvement, turnaround times to create these furnished geometric representations for a single layout can be upwards of one week [16]. An alternative to creating the scene from scratch is instead using Matterport scans [26] as the base layer, a technique utilized by the virtual staging tool Roomy [41]. Structural scans, like those from Matterport and Occipital [30], require initial involvement by the user to scan the space, thus adding a constraint at the beginning of the pipeline.

As our work is focused on providing out-of-the-box consumer utility, we address the aforementioned turnaround time constraints by automatically capturing elements of lighting and the space to generate imagery with minimal human labor at a comparable level of realism.



Figure 2: Low LOD 3D model as displayed in an AR scene (left). Fused Photo render of the same model composited into a reconstructed scene with a high LOD model (center, right).

3 FUSED PHOTO

Fused Photo is an AR system that lets customers obtain photorealistic renders of furniture composited in their space by capturing sufficient information from the user’s context to recreate a scene in an offline rendering service that replaces low LOD 3D models with high LOD analogues. Our implementation is decomposed into two parts: an ARKit client that captures context from the user’s environment and a server that interprets the payload to recreate the scene in an offline 3D rendering engine to produce the final image.

3.1 Scene Capture

In this section we describe the mechanisms employed by the AR client to capture the user’s context for the generation of a Fused Photo. This includes the selection and positioning of the products that will be staged, the environmental backdrop, topological information of the scene, and the lighting conditions of the environment.

3.1.1 Topology. In order to accurately place objects at the correct scale and perspective, we need an understanding of the user’s space. We found that reducing the scene to a collection of primary planes was sufficient to accurately stage products, provided they are placed either on the floors or hung on walls. We rely on ARKit’s plane finding capabilities to capture vertical and horizontal planes by having the user move their phone around in an initial capture stage.

3.1.2 Proxy Model Placement. Once the initial sequence of planes is captured, the customer is able to add, translate, and rotate 3D models atop them. These models are low LOD with simple materials and low polygon counts. In essence, these proxy models are far from an accurate representation of the product, but serve as a visual guide for the user to stage the scene, giving them a sense of the

spatial features of the product and its location in space. The models used to develop Fused Photo come from Wayfair’s catalog, which can be accessed through their 3D Model API [42].

3.1.3 Environmental Backdrop. When the user has finished staging a proxy model, they are prompted to take a picture. By taking the photo, the user stops the AR feed, but they can continue making adjustments to the products within this frame. This picture represents the slice of the user’s environment that the Fused Photo server will reconstruct and render in high resolution. Users can also capture a panoramic slice by panning the phone from left to right. During this pan, data from the gyroscope is utilized to capture the total field of view in degrees as well as the corresponding photo frame for every seven degrees to ensure enough overlap for panoramic stitching. OpenCV’s built-in stitching is then employed to stitch together the panoramic backdrop.

3.1.4 Lighting Conditions. ARKit provides the ambient light intensity in Lumen and the color temperature of the scene’s lighting in Kelvin on a per-frame basis inside the AR session. We also include the total exposure value of the scene measured in EV and the white balance point for the captured frame in the camera parameters sent from the client. For directional light sources, we devised a simple strategy, where users tap on the screen, giving the service the position of lights in the scene. We raycast from the user’s tap to obtain the estimated light locations.

3.2 Scene Reconstruction

The render thread’s job is to recreate the scene in an offline 3D rendering software based on the contextual data captured by the AR client. The reconstructed scene must generate a rendered image that matches the paused AR camera feed, replacing the low resolution 3D model of the product with a high resolution version (see Figure 3). Instead of recreating a 3D scene each time a request from the client is received, we created a template scene that is modified each time a new set of parameters is received. The following section describes how the scene is modified to generate a Fused Photo.

3.2.1 Model Placement. The AR model transforms are included in the payload from the client. For every model placed in the AR scene that within the view frustum of the camera when the feed is stopped, the render thread imports an analogous high resolution

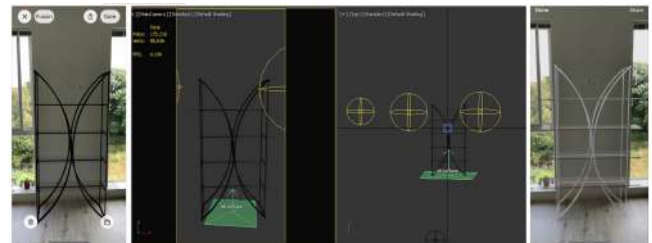


Figure 3: Information about planes in the scene, the position of the low LOD model, the position of lights, and planes detected by the AR framework are transmitted to the service, which reconstructs the scene in 3D rendering software to create a photorealistic composite.

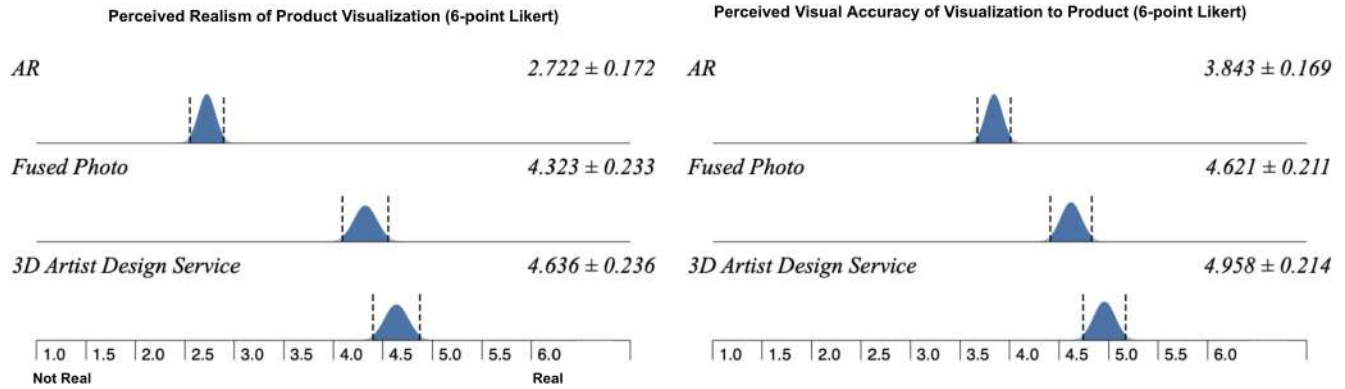


Figure 4: (Left) Results from preliminary study comparing participants perception of realism of a product visualized in AR, Fused Photo, and renders generated by 3D artists on a 6-point Likert scale. (Right) Same mediums compared in their ability to accurately depict the appearance of a product on a 6-point Likert Scale.

model from a repository. The model is then inserted in the scene by applying the world transformation which scales, rotates, and positions the object.

3.2.2 Camera Matching. The template scene contains a single virtual camera that renders the composite. This camera must match the vertical and horizontal field of view of the AR camera of the client to match the perspective view of the captured image. These parameters can be obtained from the AR camera and mapped directly onto the virtual camera to achieve the desired effect. Exposure parameters must be set directly based on the exposure offset of the last frame captured from the AR session. We match the global exposure offset of the scene to this value to ensure the light estimation from the AR scene holds in our render. The transform of the camera is matched by employing the world transform of the AR camera obtained from the client.

3.2.3 Environment. The planar image captured by the AR client must be modified to be used as an environment in the offline 3D scene. 3D rendering software allows for the application of a 360 environment map to relight the scene and render realistic reflections on objects in a render. Given the known vertical and horizontal field of views of the ARKit camera, we can project the image onto a section of a unit sphere that represents the captured field of view. The warped image can be mapped onto a equirectangular representation such that, when applied in the offline 3D scene and rendered with a camera with the appropriate field of view, the original unwarped image is yielded.

3.2.4 Lighting. The render thread must recreate the lighting in the scene based on the overall ambient light intensity, the overall light temperature, and the positions of prominent sources of lighting in the scene. Therefore, in the 3D scene we employ one ambient light component and several point lights to represent the light positions specified by the user. Given that the user's selection might not represent an exhaustive list of lighting sources in the environment or the contribution of each light to the overall luminosity of the

scene, we approximate by dividing the total intensity obtained from ARKit equally amongst directional and ambient light sources.

3.2.5 Shadows. In order to capture the effect of the lighting on the environment and the products placed in the scene, we employ the planar information captured by the client. For each plane detected in the AR session, we create a matte plane that captures shadows and bounces lights like a real surface. The positions and orientations of the planes are matched the same way the camera and 3D models are placed in the scene.

4 RESULTS

We conducted a within subject online survey with 33 participants to understand how Fused Photos compare to other virtual product visualization techniques employed in online retail. Three types of visualization experiences were tested with images selected to represent different experiences in the market for a total of 23 images. The first were AR visualization tools employed by online retailers from Houzz (4 images) [19], Wayfair's View-in-Room 3D (4 images) [46], and IKEA Place (4 images) [44]. We also included imagery from two designer services that create product visualizations: Roomy (2 images) [41] and Modsy (3 images) [27]. Finally, we created six Fused Photos in different environments using the system presented in this paper.

Each survey question presented an image that contained a product composited onto a real environment. Participants were instructed to determine how real the composited object appeared within the environment on a 6-point Likert scale. Furthermore, a picture of the actual product was included in a follow up question, asking the participant to determine if the visualization of the product accurately portrayed the visual appearance of the real product on a 6-point Likert scale.

A one-way ANOVA between subjects was conducted to compare the realism of each medium and its capability to accurately portray the visual characteristics of a product (see Figure 4). There was a significant effect found both in terms of realism of the medium [$F(2,$



Figure 5: Fused Photos generated in different lighting environments, showcasing different products composited in the customer's environment.

756)=104.06, $p<0.001$] and the ability to portray visual characteristics of a product [$F(2, 756)=34.38$, $p<0.001$]. Post-hoc Tukey tests indicate that Fused Photos ($M=4.31$, $SD=0.23$) and designer services ($M=4.64$, $SD=0.24$) are perceived significantly more real than other AR experiences ($M=2.72$, $SD=0.17$) currently employed in commercial online retail ($p<0.01$). Moreover, no statistical significance was found between the realism achieved by Fused Photos and manually staged composites ($p>0.05$). Fused Photos ($M=4.62$, $SD=0.21$) and 3D artist renders ($M=4.96$, $SD=0.21$) were also considered to be significantly more capable of accurately representing the visual appearance of the real product than the AR group ($M=2.84$, $SD=0.17$, $p<0.01$), with no significant difference between Fused Photos and manual design work ($p>0.05$).

5 DISCUSSION

The captured Fused Photos show that we are able to accurately recreate the imagery taken by the ARKit client in the remote rendering service (Figure 5). By matching camera field of views and leveraging the spatial relationships of objects in the AR scene, we can effectively match the perspective view while swapping the low resolution model by a high LOD analogue in the same position.

The survey results support the claim that the Fused Photo system presented in this work can bridge the gap between the virtual and physical visualizations of a product, achieving more realism than traditionally employed mechanisms in retail. Moreover, our system is able to generate imagery comparable to current designer services in the market that employ manual 3D artist work with turnaround times in the order of days/weeks. This suggests that the use of Augmented Reality as a tool for staging high resolution renders can improve the visual quality of products composited in the user's space and reduce the perceptual gap between what the customer sees online and what they get.

The Fused Photo system offers an asynchronous AR paradigm, where the user must wait approximately ten seconds to obtain the high quality render results. Although this can be seen as a shortcoming, once the user creates a Fused Photo, they can manipulate the

3D object within the scene or even swap it for a different product, allowing the user to make use of the scene even after they have left the space. This makes the generated content more reusable than traditional AR feeds and more accessible to the demographics of online shoppers of home goods. Moreover, the generated Fused Photo scene can be shared with others (such as a designer service) who can stage the space without having to be present, further supporting that a real time paradigm is not required for a meaningful interactive experience.

We note that, depending on the process by which the real-time model was generated there may be some discrepancies between the sizes of the low and high LOD versions. These incongruities are minimal and hard to notice unless the images are compared side by side. The resulting imagery is able to capture the likeness of the actual product much more accurately than the real-time model, due to the quality of textures, color, and topology of the mesh being truer to the original object. It is worth noting that this effect is more pronounced for certain materials such as cloth, glass, and metallic surfaces, while less pronounced for materials such as plastic and wood. Moreover, the photorealistic quality of the Fused Photo render is also determined by the quality of the high LOD model, where a poorly modeled product would still look fake regardless of the accuracy of the Fused Photo system in recreating the scene. Beyond the appearance of the model being of higher quality, Fused Photos are able to recreate lighting conditions that help the product blend with the environmental backdrop, despite a very simple approximation technique. The invisible virtual planes capture shadows that look truer to life than those portrayed in existing real-time AR frameworks, including self-shadowing.

6 CONCLUSION

In this paper we presented a new system that leverages Augmented Reality to generate photorealistic imagery by capturing context from the user's environment to recreate a composite scene using an offline rendering engine. This technique enables online retailers of home goods to use low LOD models for in-situ staging as a proxy,

which the remote rendering service then replaces with high LOD counterparts, reducing the gap between the physical product and what the customer visualizes in the online shopping experience. We have described the techniques employed by our AR client to capture scene from the user's context, and how the remote server leverages this information to recreate a scene in offline 3D rendering software to render a Fused Photo with high resolution models swapped in. Our results showed that Fused Photos outperform the real-time AR visualization techniques both in realism and ability to represent a product's visual features more accurately, at the cost of additional rendering time in the order of seconds, with minimal additional user interaction. Furthermore, we have shown how the output of our system is comparable to renders staged by 3D artists in terms of realism, but done in a fraction of the time.

7 FUTURE WORK

Although the visual quality of the proposed system was evaluated, the interactive elements still require formal evaluation to understand how Fused Photos compare to traditional AR experiences in terms of usability, and whether the render time of the Fused Photo system is a deterrent to users. In future iterations of this work, we also hope to incorporate real-time ray tracing (RTX) to achieve near real-time interactions with the Fused Photo rendering service, bridging the interactivity gap between traditional AR and our solution. In terms of the quality of the visualization, future iterations should incorporate the existing body of work on light estimation to eliminate the need for human input in the relighting process. Finally, as LIDAR-equipped devices are becoming more ubiquitous, we hope to explore how a fine-grained capture of the scene (e.g. point clouds) can be integrated with the Fused Photo experience to enhance visual coherence of shadows and occlusion within the user's context.

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